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Assignment Name : Question Paper Prelim

Q1] 1) List the application of AI

→ Autonomous Systems, E-Commerce, Healthcare

2) Concept of Datawarehousing

→ Datawarehousing is a concept in which the raw data is gathered and organize it into a structured format.

3) Define search strategy

→ A search strategy refers to the systematic approach used in Data Mining to discover hidden patterns, correlations or anomalies within large datasets.

4) What are the techniques of AI

→ 1) Search (Defining AI problem as a state space search) initial state to final state (desirable & undesirable)

2) Use of knowledge - once the search strategy is defined, the system must use Domain knowledge to make decisions.

3) Abstraction - is a process of removing unnecessary details to focus on core logic of problem

5) Define Expert System

→ Expert system is a computer Application that mimics the decision-making ability of a human expert. It is designed to solve complex problems by reasoning through bodies of knowledge, rather than traditional procedural code

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6) Write about hill climbing algorithm

→ Hill climbing algorithm is a heuristic search algorithm used in AI to find optimal solution. It starts with an initial solution & continuously moves to a neighboring solution that has a better value. The process continues until no better neighbour is found, reaching a peak

7) Write about generate & test

→ It is simple problem-solving technique in AI. In this method possible solutions are generated by one by one & then tested to check whether they satisfy the given problem condition. The process continues until correct solution is found

8) Define ETL process

→ 1) E (Extraction) - In this, all the raw data is collected & stored in one place

2) Transform - In this, the raw data is "dirty", so then it is cleaned and put it in std format

3) Load - Then, the data is save in database

9) ~~Define~~ OLTP & OLAP

→ OLTP (Online Transaction processing) is used to manage day-to-day transactions such as data entry & deletions
eg - banking systems, billing system

OLAP - Online Analytical processing is used for data analysis & decision-making. It processes large volumes of historical data to generate reports.

Q2] →

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Q2] 1) Explain advantages of Datawarehousing

→ Datawarehouse means the data is collected then reports are generated on it. Data-analysis is also done here

1) Integrated Data - A datawarehouse combines data from multiple sources (database, application, etc) into single, unified system, making it easier to access & manage

2) Better Decision Making - It provides accurate & consistent data, helping managers & organizations make informed & strategic decisions

3) Historical Data analysis - Data warehouse store large amt of historical data, allowing users to analyze trends, pattern over time

4) Improved Query performance - They are optimized for complex queries & reporting, so users can retrieve information quickly without affecting daily operations

5) Data Consistency - Data is cleaned & transformed before storing, ensuring reliable & high-quality information

6) Supports Business Intelligence (BI) - Datawarehousing works with BI tools to generate reports, dashboards, & visualization for better insights

2) Explain the star schema technique of modelling a datawarehouse

→ Star schema is a data modeling technique used in data warehouse where a central fact table is connected to multiple dimension table, forming a structure that looks like a star

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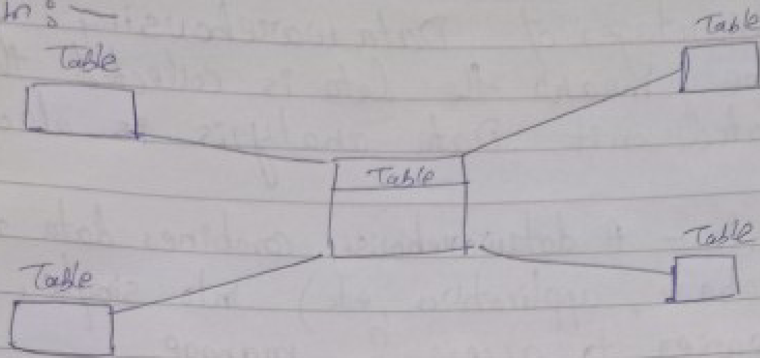
Diagram :—



Component :—

1) Fact table – Contains numerical data like sales, profit. Include the foreign keys that link to dimension tables

2) Dimension Tables – contain descriptive information such as customer name, product, time, etc

The fact table is place at center & dimension tables are connected directly to it.

Advantages :—

Simple & easy to understand

Faster query performance

Suitable for reporting & analysis

3) State ~~4~~ component with the help of which problem can be well formulated.

→ 1) Initial state

The IS represents the starting condition of problem. It defines where the problem-solving process begins. For eg – In puzzle it is initial arragement of pieces before any moves

2) State space

The SS includes all possible states that can be reached from the IS by appling different actions

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It represents the complete set of situation that may occur while solving the problem

3) Operators (Action) - They are the rules that define how one state can be transformed into another. These actions help in moving step-by-step from the initial state towards the goal state.
For eg - moving a tile in a puzzle

4) Goal state - It is the desired final outcome of the problem. It specifies the condition that must be achieved to consider the problem solved. The objective of problem-solving process is to reach this state from initial state using valid operators.
This 4 components help in clearly defining & solving a problem in systematic & efficient way

4) Difference between OLTP & OLAP system

→

	OLTP	OLAP
1.	Handles the day-to-day transactions	Used for data-analysis & decision making
2.	works with current, detailed data	works with historical, data
3.	Online Transaction processing	Online Analytical Processing
4.	Insert, update, delete	Complex queries analysis
5.	very fast for transactions	optimized for query performance
6.	eg - Banking system	eg - DWH Business Intelligence

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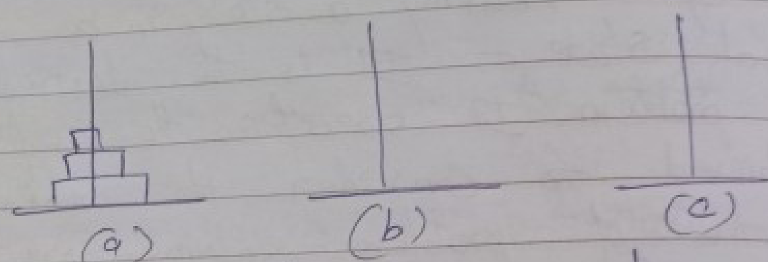
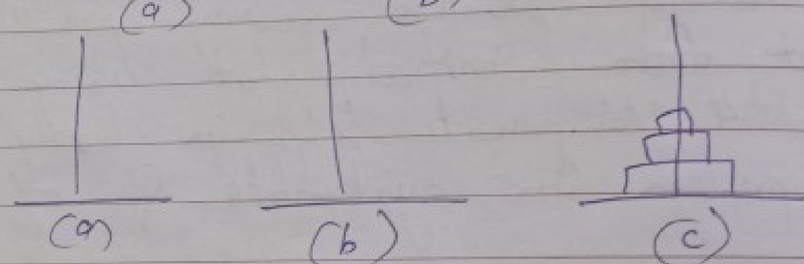
5) Give the state space representation for "Tower of Hanoi problem". Q3

→ Problem description : The Tower of Hanoi consists of 3 peg (A, B, C) and n disks of different size. The disk are initially stacked on one peg in decreasing size order

Initial state :-



Goal state :



Moves :-

1) Move one disk at a time
2) Only the top disk can be moved
3) A larger disk cannot be placed on smaller disk

Initial state - all disks are placed on same peg (a) in proper order
Goal state - all disks must be moved to destination (c) in same order

State space :- The state space consists of all possible valid arrangement of disks across the 3 peg following the rules.

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Q3] 1) Write application of Data Mining

→ Data Mining is the process of discovering patterns, correlations & with large datasets to predict outcome

1) Marketing & Retail (Market Basket Analysis) – Retailer use data mining to identify which products customers frequently buy together. For eg – data show people buy bread & butter together so store can place & offer discount.

2) Banking & finance – Bank uses data mining to identify "outlier" transactions that differ from user's typical spending habits Helps in fraud detection.

3) Healthcare & Medicine – Data mining helps doctors & researchers identify patterns in patient symptoms to provide more accurate diagnoses

Disease prediction – predicting the likelihood of patient developing a condition like diabetes based on their family history & lifestyle

4) Telecommunications — Telecom companies deal with millions of calls & data packets daily. They use data mining for churn prediction and network optimization

Churn – identifying customers who are likely to switch to a competitor so the company can offer them a better deal to stay

network optimization – finding peak usage times to manage network traffic & prevent dropped calls.

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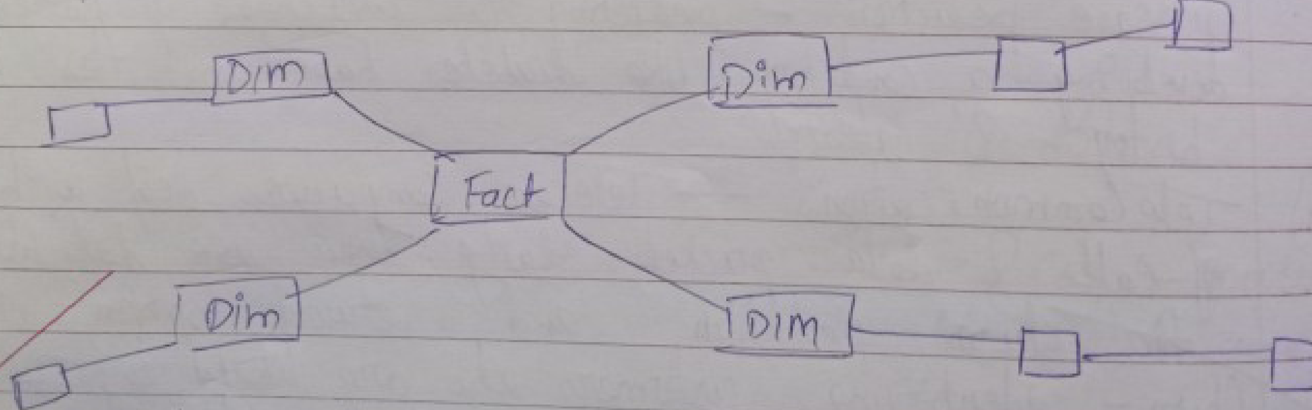
2) Explain the details snowflake schema

→ In data warehousing, a snowflake schema is an extension of the star schema where the diagram resembles a snowflake. It is used to organize data in a way that reduce redundancy

- Definition — It consist of one centralized fact Table connected to multiple Dimension Tables. Unlike a star Schema this dimension table are normalized, meaning they are broken down into further sub-dimension tables

The primary characteristic is that the dimensions are expanded into multiple related tables. For eg - instead of having "city" & "country" in one "location" table, the "location" table might only have a "city-ID" when links to separate "city" table, links to "country".

Diagram :—



Fact Table - contains numerical value like sales, amt, quantity, profit.

Dimension Table — Describes the data in Fact table & they are broken into multiple related tables.

3)

→

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3) Explain with eg - the concept of linear regression

→ Linear regression is a statistical method used to find the relationship betw 2 variables by fitting a straight line. It helps in predicting the value of ~~b~~ one variable based on another variable

For eg - $y = mx + c$

hours studied (x)	marks (y)
1	40
2	50
3	60
4	70

When we plot these points, they form a straight line, using linear regression, we can draw a line that best fits

eg - If a student studies 5 hours, the model may predict around 80 marks

Linear regression is a simple & powerful technique widely used for prediction & trend analysis, used in fields like business, economics & machine learning

4) Explain the concept of fact table & process of ETL

→ The Fact table is the central table in data warehouse that stores quantitative (numerical) data about business processes

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Key features :—

Q4

- Contains measures like sales amt, profit, quantity
- Includes foreign keys linking to dimension tables

eg —

Sales Fact Table :—

Product ID	Customer ID	Time ID	Sales amt ID

Sales — Fact Table

Product ID — Dimension Table

Time ID

ETL (Extract, ~~Load~~ Transform, Load)

1) Extract — data is collected from multiple sources (databases, files, APIS)
eg — sales data from different branches

2) Transform — Data is cleaned, formatted & processed
Removes errors, duplicates & converts data into proper format

3) Load — Processed data is stored into data warehouse. Ready for analysis & reporting

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Q4) Answer the following :-

1) Explain concept of spark technology

→ Apache spark is defined as an open source, distributed computing system used for big data processing. It was designed to overcome the limitations of the older MapReduce model by providing significantly faster data processing

Four piller :-

1) High Speed In Memory Processing
The most important concept of spark is in-Memory computing. Unlike Hadoop MapReduce, which writes data to the physical disk after every step, spark keeps data in the RAM (Random Access Memory)
Result — This makes spark up to 100 times faster for certain application, especially iterative algorithms used in ML.

2) RDD (Resilient Distributed Dataset) - The RDD is the fundamental data structure of spark.

- Resilient - It is fault-tolerant; if a part of da[ta] is lost, spark can rebuild it using "Lineage" (a record of how the data was created)
- Distributed - The data is divided into chunks &
- spread across multiple computers (nodes) in a cluster

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- Dataset – It is a collection of objects that can be processed in parallel.

3) Lazy Evalution
spark does not exeute operations immediately. Instead, it records the transformations you want to perform on the data in a DAG (Directed Acyclic Graph). The exeution only starts when an Action is called.

4) The unified stack
Spark is not just a single tool, it is a unified Engine that supports various types of data processing in one place:
spark SQL – For working with structured data using SQL queries
Spark streaming – For processing real-time data
Mllib – A library for Machine learning

2) Explain in detail components of data warehouse

→ Datawarehouse (DW) is a specialized database used for reporting & data analysis. Unlike a std database, it is "subject - oriented" & "integrated".

1) Data Sources (operational systems)
This is the "input" layer. A data warehouse collects data from various places:

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2) Unified Data Access
It provides a common way to access a variety of data sources, including Hive, Avro, Parquet & JDBC. This function ensures that you can join data process different formats using single interface without needing to write specialized code for every source.

3) Hive Compatibility
Spark SQL is designed to be compatible with Apache Hive. It can run unmodified Hive queries on existing Hive warehouses & access Hive UDFs & metadata. This makes it an ideal tool for migrating or extending legacy Hadoop workflows.

4) Standard Connectivity (JDBC / ODBC)
Spark SQL includes a server mode with standard JDBC & ODBC connectivity. This allows business intelligence (BI) tools (like Tableau, Power BI or) & other external applications to connect to the spark engine just like they would to a traditional relational database.

4] Explain the process of data cleaning process in data mining

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→ The Data cleaning process (often called as Data scrubbing) is a crucial step in Data Preprocessing. Its main goal is to improve data quality by removing "noise" & inconsistencies

4 process :—

1) Handling missing values

This step addresses data records where certain attributes are blank. You can handle this by :-

- Ignoring the tuple — usually done if the class label is missing
- Filling manually — Manually entering the data missing
- Using a global constant — filling blanks with "unknown" or 0.
- Using a measure of central tendency. Filling with the mean or median of attribute

2) Noisy Data Smoothing

"Noise" refers to random error or variance in a measured variable. To "smooth" out this noise, several technique are used :

- Binning — sorting data into small "bins" and smoothing by bin means or boundaries
- Regression — fitting the data to a linear or multiple regression function

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3) Identifying & removing inconsistencies
Data often comes from multiple sources, leading to inconsistencies. This step involves :-
- Correcting Manual Entry Errors - fixing types (eg - "M" vs "Male").
- Standardizing formats - Ensuring dates are all in the same format (DD-MM-YYYY).
- Handling Redundancy - Removing duplicate records that may have been integrated from different databases.

4) Data Transformation
— While sometimes viewed as a separate step, cleaning often requires transforming data to make it usable.
- Normalization - Scaling numerical data to a specific range (like 0 to 1).
- Generalization - Replacing low-level "primitive" data with higher level concepts (eg - replacing "city" with "state")

5] Explain the advantages of AI in real world.

→ Artificial Intelligence (AI) has moved from theoretical research to a core component of modern infrastructure.

~~Ex~~:

1) Automation of Repetitive Tasks.
AI excels at handling mundane, high-volume tasks that would otherwise consume human time. By

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automating data entry, email sorting or manufacturing assembly lines, AI allows human to fows on creative & high-level decision-making. This leads to increased productivity & lower operational costs for businesses.

2) Enchanced Accuracy & Precision
Unlike humans, AI models donot suffer from fatigue or emotional distraction. In fields like healthcare, AI algorithms can scan medical images to detect anomalies with a level of precision that rivals or exceeds specialists. In finance, AI reduces "human error" in accounting & fraud detection by spotting patterns in millions.

3) 24/7 Availability & support
AI- powered system, such as Chatbots & Virtual Assistants (Like Alexa, siri) provide "always-on" service. They can handle customer queries, provide technical support, or monitor security system at any time of day or night without needing breaks. This ensures that services remain accessible to users across different time zones.

4) Data-Driven Decision Making
AI can process & analyze "Big Data" far faster than my human. By identifying hidden trends & patterns,

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AI helps in predictive Analytics. For eg:-

- In Agriculture - Predicting crop yields based on weather patterns
- In logistics — optimizing delivery routes to save fuel & time
- In Retail - Suggesting products to customers based on their browsing history

Q5] Short note :—

1) Explain in detail BFS & DFS algorithm.

→ BFS & DFS are the two most fundamental algorithm used to transverse or search through nodes in a graph or tree

1) BFS (Breadth - First - Search)

BFS explores the neighbor nodes first, before moving to the next level neighbours. It transverse the graph layer by layer. (horizontally)

- Data Structure — uses a Queue (FIFO)
- Mechanism = It starts at a ~~node to~~ root node, visits all its direct neighbors, then moves to the neighbors of those
- Use Case - Ideal for finding the shortest path in an unweighted graph.

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2. Algorithm steps
Evaluate the initial state - if the goal state, quit & return success
Loop unitil a solution is found or no new operator can be applied :
- select a new operator & apply it to the current state to generate a neighbour state.
- Compare : Evaluate the new neighbor state
- Move : If the neighbor state is better than the current state, make the neighbor state the current state.
- stay : If the neighbor state is not better, ignore it & continue searching other neighbor

3. Characteristics :-
Greedy Approach - It only looks at immediate neighbor nodes & doesn't keep track of path taken.
Memory Efficient - It only stores the current state & the objective function value

3] Explain mean & end analysis with eg:-

→ Means - Ends Analysis is a strategy used in problem - solving agents to reduce the difference between the current state & goal state. Instead of searching blindly, it identifies the obstacles preventing the goal & creates "sub-goals" to remove them one by one.